

Hsu, Chun-Yuan

 My Portfolio |  chyhsu@umich.edu |  Chyhsu |  Chun-Yuan Hsu

Skills

- Languages:** Python, C++, Go
- AI & ML:** PyTorch, AI Agents, Claude Agent SDK, LLM Integration (AWS Bedrock), RAG (ChromaDB), Embeddings, Semantic Search
- Cloud & DevOps:** Docker, Kubernetes, AWS, GCP, GitLab CI/CD, Prometheus, Elastic Stack (ELK), Alertmanager, Grafana, NATS
- Frameworks/Systems:** Node.js, React, Linux (Debian, Arch), Git, Scrum

Education

- Master of Science in Data Science**, University of Michigan – Ann Arbor, MI, USA Sep 2025–Present
- Master of Science in Computer Science**, National Tsing Hua University – Hsinchu, Taiwan Sep 2022–Jan 2025
- Bachelor of Science in Civil Engineering**, National Cheng Kung University – Tainan, Taiwan Sep 2018–Jun 2022

Experience

- Digital Workflow Development Department Intern, TSMC, Hsinchu, Taiwan** May 2026 – Present
- Developed an AI-agent workflow with the Claude Agent SDK to automatically triage backend alerts and generate structured incident-analysis reports.
 - Built integrations to ingest alerts from Alertmanager and retrieve logs and metrics from Kubernetes workloads via ELK and Prometheus.
 - Designed a hypothesis-driven investigation loop that correlates alerts, logs, and metrics to identify likely root causes and summarize actionable findings for engineering teams.
- Backend R&D Internship, QNAP, Taiwan** Jan 2025 – Jul 2025
- Built a retrieval-augmented Jira issue search system using AWS Bedrock and ChromaDB embeddings, increasing developer issue-resolution efficiency by 50%.
 - Developed an MCP-based Jira search server that integrates with IDEs, enabling developers to query and explore issues directly from their coding workflow.
 - Refactored Device Avatar microservices from Python to Go, achieving a 30% performance gain and optimizing deployment on Kubernetes.
 - Diagnosed and patched a critical memory leak in cloud production by correlating Grafana metrics with execution traces.

Projects

- Toward Interpretable Brain Age Prediction and AD Classification**  report
- Built an interpretable neuroimaging pipeline for brain age regression and Alzheimer’s Disease (AD) classification using structural MRI scans from OpenBHB and ADNI.
 - Adapted NeuroVFM, a pretrained 3D vision transformer, to extract patch-level MRI embeddings and aggregate variable-length features with attention-based multiple instance learning.
 - Trained joint classification and regression heads, reaching 0.873 diagnostic accuracy, 0.775 macro F1, 3.54-year MAE, and 0.966 R^2 while mapping attention back to clinically relevant brain regions.

Lilac

- Designed and implemented a cross-cloud IaC lifting framework that automatically reconstructs Terraform configurations from existing deployments (“brownfield” infrastructure) across Azure, Google Cloud, and AWS.
 - Developed a pipeline that integrates Large Language Models (LLMs) with symbolic verification to learn resource–dependency mappings from cloud APIs.
 - Conducted large-scale evaluation on real cloud environments, achieving higher accuracy and coverage than existing tools (e.g., Terraformer) while maintaining correctness.
- VizThinker** 
- Created an innovative graph-based interface for interacting with LLMs, reimagining traditional linear chatboxes into a visual conversation graph.
 - Implemented branching and node-based history navigation to support complex idea exploration.
 - Deployed on Google Cloud Platform with a modern stack of Node.js, React, and Python.